Speex Reference Manual

1.0

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Chapter 1

Speex Compound Index

1.1 Speex Compound List

Here are the classes, structs, unions and interfaces with brief descriptions:

SpeexBits 5
SpeexCallback 7
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Chapter 2

Speex File Index

2.1 Speex File List

Here is a list of all documented files with brief descriptions:

speex.h (Describes the different modes of the codec) 13
speex_bits.h (Handles bit packing/unpacking) 23
speex_callbacks.h (Describes callback handling and in-band signalling) 27
speex_header.h (Describes the Speex header) 30
speex_stereo.h (Describes the handling for intensity stereo) 32

Chapter 3

Speex Class Documentation

3.1 SpeexBits Struct Reference

`#include <speex_bits.h>`

Public Attributes

- char * bytes
- int nbBits
- int bytePtr
- int bitPtr
- int owner
- int overflow
- int buf_size
- int reserved1
- void * reserved2

3.1.1 Detailed Description

Bit-packing data structure representing (part of) a bit-stream.

3.1.2 Member Data Documentation

3.1.2.1 int SpeexBits::bitPtr

Position of the bit "cursor" within the current byte

3.1.2.2 int SpeexBits::buf_size

Allocated size for buffer

3.1.2.3 int SpeexBits::bytePtr

Position of the byte "cursor"

3.1.2.4 char* SpeexBits::bytes

"raw" data

3.1.2.5 int SpeexBits::nbBits

Total number of bits stored in the stream

3.1.2.6 int SpeexBits::overflow

Set to one if we try to read past the valid data

3.1.2.7 int SpeexBits::owner

Does the struct "own" the "raw" buffer (member "bytes")

3.1.2.8 int SpeexBits::reserved1

Reserved for future use

3.1.2.9 void* SpeexBits::reserved2

Reserved for future use

The documentation for this struct was generated from the following file:

- speex_bits.h

3.2 SpeexCallback Struct Reference

`#include <speex_callbacks.h>`

Public Attributes

- int callback_id
- speex_callback_func func
- void * data
- void * reserved1
- int reserved2

3.2.1 Detailed Description

Callback information

3.2.2 Member Data Documentation

3.2.2.1 int SpeexCallback::callback_id

ID associated to the callback

3.2.2.2 void* SpeexCallback::data

Data that will be sent to the handler

3.2.2.3 speex_callback_func SpeexCallback::func

Callback handler function

3.2.2.4 void* SpeexCallback::reserved1

Reserved for future use

3.2.2.5 int SpeexCallback::reserved2

Reserved for future use

The documentation for this struct was generated from the following file:

- speex_callbacks.h

3.3 SpeexHeader Struct Reference

`#include <speex_header.h>`

Public Attributes

- char speex_string [8]
- char speex_version [SPEEX_HEADER_VERSION_LENGTH]
- int speex_version_id
- int header_size
- int rate
- int mode
- int mode_bitstream_version
- int nb_channels
- int bitrate
- int frame_size
- int vbr
- int frames_per_packet
- int extra_headers
- int reserved1
- int reserved2

3.3.1 Detailed Description

Speex header info for file-based formats

3.3.2 Member Data Documentation

3.3.2.1 int SpeexHeader::bitrate

Bit-rate used

3.3.2.2 int SpeexHeader::extra_headers

Number of additional headers after the comments

3.3.2.3 int SpeexHeader::frame_size

Size of frames

3.3.2.4 int SpeexHeader::frames_per_packet

Number of frames stored per Ogg packet

3.3.2.5 int SpeexHeader::header_size

Total size of the header (sizeof(SpeexHeader))

3.3.2.6 int SpeexHeader::mode

Mode used (0 for narrowband, 1 for wideband)

3.3.2.7 int SpeexHeader::mode_bitstream_version

Version ID of the bit-stream

3.3.2.8 int SpeexHeader::nb_channels

Number of channels encoded

3.3.2.9 int SpeexHeader::rate

Sampling rate used

3.3.2.10 int SpeexHeader::reserved1

Reserved for future use, must be zero

3.3.2.11 int SpeexHeader::reserved2

Reserved for future use, must be zero

3.3.2.12 char SpeexHeader::speex_string[8]

Identifies a Speex bit-stream, always set to "Speex "

3.3.2.13 char SpeexHeader::speex_version[SPEEX_HEADER_VERSION_LENGTH]

Speex version

3.3.2.14 int SpeexHeader::speex_version_id

Version for Speex (for checking compatibility)

3.3.2.15 int SpeexHeader::vbr

1 for a VBR encoding, 0 otherwise

The documentation for this struct was generated from the following file:

- speex_header.h

3.4 SpeexMode Struct Reference

`#include <speex.h>`

Public Attributes

- void * mode
- mode_query_func query
- char * modeName
- int modeID
- int bitstream_version
- encoder_init_func enc_init
- encoder_destroy_func enc_destroy
- encode_func enc
- decoder_init_func dec_init
- decoder_destroy_func dec_destroy
- decode_func dec
- encoder_ctl_func enc_ctl
- decoder_ctl_func dec_ctl

3.4.1 Detailed Description

Struct defining a Speex mode

3.4.2 Member Data Documentation

3.4.2.1 int SpeexMode::bitstream_version

Version number of the bitstream (incremented every time we break bitstream compatibility

3.4.2.2 decode_func SpeexMode::dec

Pointer to frame decoding function

3.4.2.3 decoder_ctl_func SpeexMode::dec_ctl

ioctl-like requests for decoder

3.4.2.4 decoder_destroy_func SpeexMode::dec_destroy

Pointer to decoder destruction function

3.4.2.5 decoder_init_func SpeexMode::dec_init

Pointer to decoder initialization function

3.4.2.6 encode_func SpeexMode::enc

Pointer to frame encoding function

3.4.2.7 encoder_ctl_func SpeexMode::enc_ctl

ioctl-like requests for encoder

3.4.2.8 encoder_destroy_func SpeexMode::enc_destroy

Pointer to encoder destruction function

3.4.2.9 encoder_init_func SpeexMode::enc_init

Pointer to encoder initialization function

3.4.2.10 void* SpeexMode::mode

Pointer to the low-level mode data

3.4.2.11 int SpeexMode::modeID

ID of the mode

3.4.2.12 char* SpeexMode::modeName

The name of the mode (you should not rely on this to identify the mode)

3.4.2.13 mode_query_func SpeexMode::query

Pointer to the mode query function

The documentation for this struct was generated from the following file:

- speex.h

3.5 SpeexStereoState Struct Reference

`#include <speex_stereo.h>`

Public Attributes

- float balance
- float e_ratio
- float smooth_left
- float smooth_right
- float reserved1
- float reserved2

3.5.1 Detailed Description

State used for decoding (intensity) stereo information

3.5.2 Member Data Documentation

3.5.2.1 float SpeexStereoState::balance

Left/right balance info

3.5.2.2 float SpeexStereoState::e_ratio

Ratio of energies: E(left+right)/[E(left)+E(right)]

3.5.2.3 float SpeexStereoState::reserved1

Reserved for future use

3.5.2.4 float SpeexStereoState::reserved2

Reserved for future use

3.5.2.5 float SpeexStereoState::smooth_left

Smoothed left channel gain

3.5.2.6 float SpeexStereoState::smooth_right

Smoothed right channel gain

The documentation for this struct was generated from the following file:

- speex_stereo.h

Chapter 4

Speex File Documentation

4.1 speex.h File Reference

Describes the different modes of the codec.

`#include "speex_bits.h"`

Compounds

- struct SpeexMode

Defines

- #define SPEEX_SET_ENH 0
- #define SPEEX_GET_ENH 1
- #define SPEEX_GET_FRAME_SIZE 3
- #define SPEEX_SET_QUALITY 4
- #define SPEEX_GET_QUALITY 5
- #define SPEEX_SET_MODE 6
- #define SPEEX_GET_MODE 7
- #define SPEEX_SET_LOW_MODE 8
- #define SPEEX_GET_LOW_MODE 9
- #define SPEEX_SET_HIGH_MODE 10
- #define SPEEX_GET_HIGH_MODE 11
- #define SPEEX_SET_VBR 12
- #define SPEEX_GET_VBR 13
- #define SPEEX_SET_VBR_QUALITY 14
- #define SPEEX_GET_VBR_QUALITY 15
- #define SPEEX_SET_COMPLEXITY 16
- #define SPEEX_GET_COMPLEXITY 17
- #define SPEEX_SET_BITRATE 18
- #define SPEEX_GET_BITRATE 19
- #define SPEEX_SET_HANDLER 20
- #define SPEEX_SET_USER_HANDLER 22
- #define SPEEX_SET_SAMPLING_RATE 24

- #define SPEEX_GET_SAMPLING_RATE 25
- #define SPEEX_RESET_STATE 26
- #define SPEEX_GET_RELATIVE_QUALITY 29
- #define SPEEX_SET_VAD 30
- #define SPEEX_GET_VAD 31
- #define SPEEX_SET_ABR 32
- #define SPEEX_GET_ABR 33
- #define SPEEX_SET_DTX 34
- #define SPEEX_GET_DTX 35
- #define SPEEX_GET_PI_GAIN 100
- #define SPEEX_GET_EXC 101
- #define SPEEX_GET_INNOV 102
- #define SPEEX_GET_DTX_STATUS 103
- #define SPEEX_SET_PF 0
- #define SPEEX_GET_PF 1
- #define SPEEX_MODE_FRAME_SIZE 0
- #define SPEEX_SUBMODE_BITS_PER_FRAME 1
- #define SPEEX_NB_MODES 3

Typedefs

- typedef void *(* encoder_init_func)(struct SpeexMode *mode)
- typedef void(* encoder_destroy_func)(void *st)
- typedef int(* encode_func)(void *state, float *in, SpeexBits *bits)
- typedef int(* encoder_ctl_func)(void *state, int request, void *ptr)
- typedef void *(* decoder_init_func)(struct SpeexMode *mode)
- typedef void(* decoder_destroy_func)(void *st)
- typedef int(* decode_func)(void *state, SpeexBits *bits, float *out)
- typedef int(* decoder_ctl_func)(void *state, int request, void *ptr)
- typedef int(* mode_query_func)(void *mode, int request, void *ptr)
- typedef SpeexMode SpeexMode

Functions

- void * speex_encoder_init (SpeexMode *mode)
- void speex_encoder_destroy (void *state)
- int speex_encode (void *state, float *in, SpeexBits *bits)
- int speex_encoder_ctl (void *state, int request, void *ptr)
- void * speex_decoder_init (SpeexMode *mode)
- void speex_decoder_destroy (void *state)
- int speex_decode (void *state, SpeexBits *bits, float *out)
- int speex_decoder_ctl (void *state, int request, void *ptr)
- int speex_mode_query (SpeexMode *mode, int request, void *ptr)

Variables

- SpeexMode speex_nb_mode
- SpeexMode speex_wb_mode
- SpeexMode speex_uwb_mode
- SpeexMode * speex_mode_list [SPEEX_NB_MODES]

4.1.1 Detailed Description

Describes the different modes of the codec.

4.1.2 Define Documentation

4.1.2.1 #define SPEEX_GET_ABR 33

Get Average Bit-Rate (ABR) setting (in bps)

4.1.2.2 #define SPEEX_GET_BITRATE 19

Get current bit-rate used by the encoder or decoder

4.1.2.3 #define SPEEX_GET_COMPLEXITY 17

Get current complexity of the encoder (0-10)

4.1.2.4 #define SPEEX_GET_DTX 35

Get DTX status (1 for on, 0 for off)

4.1.2.5 #define SPEEX_GET_DTX_STATUS 103

Used internally

4.1.2.6 #define SPEEX_GET_ENH 1

Get enhancement state (decoder only)

4.1.2.7 #define SPEEX_GET_EXC 101

Used internally

4.1.2.8 #define SPEEX_GET_FRAME_SIZE 3

Obtain frame size used by encoder/decoder

4.1.2.9 #define SPEEX_GET_HIGH_MODE 11

Get current high-band mode in use (wideband only)

4.1.2.10 #define SPEEX_GET_INNOV 102

Used internally

4.1.2.11 #define SPEEX_GET_LOW_MODE 9

Get current low-band mode in use (wideband only)

4.1.2.12 #define SPEEX_GET_MODE 7

Get current sub-mode in use

4.1.2.13 #define SPEEX_GET_PF 1

Equivalent to SPEEX_GET_ENH

4.1.2.14 #define SPEEX_GET_PI_GAIN 100

Used internally

4.1.2.15 #define SPEEX_GET_QUALITY 5

Get current quality setting

4.1.2.16 #define SPEEX_GET_RELATIVE_QUALITY 29

Get VBR info (mostly used internally)

4.1.2.17 #define SPEEX_GET_SAMPLING_RATE 25

Get sampling rate used in bit-rate computation

4.1.2.18 #define SPEEX_GET_VAD 31

Get VAD status (1 for on, 0 for off)

4.1.2.19 #define SPEEX_GET_VBR 13

Get VBR status (1 for on, 0 for off)

4.1.2.20 #define SPEEX_GET_VBR_QUALITY 15

Get current quality value for VBR encoding (0-10)

4.1.2.21 #define SPEEX_MODE_FRAME_SIZE 0

Query the frame size of a mode

4.1.2.22 #define SPEEX_NB_MODES 3

Number of defined modes in Speex

4.1.2.23 #define SPEEX_RESET_STATE 26

Reset the encoder/decoder memories to zero

4.1.2.24 #define SPEEX_SET_ABR 32

Set Average Bit-Rate (ABR) to n bits per seconds

4.1.2.25 #define SPEEX_SET_BITRATE 18

Set bit-rate used by the encoder (or lower)

4.1.2.26 #define SPEEX_SET_COMPLEXITY 16

Set complexity of the encoder (0-10)

4.1.2.27 #define SPEEX_SET_DTX 34

Set DTX status (1 for on, 0 for off)

4.1.2.28 #define SPEEX_SET_ENH 0

Set enhancement on/off (decoder only)

4.1.2.29 #define SPEEX_SET_HANDLER 20

Define a handler function for in-band Speex request

4.1.2.30 #define SPEEX_SET_HIGH_MODE 10

Set high-band sub-mode to use (wideband only)

4.1.2.31 #define SPEEX_SET_LOW_MODE 8

Set low-band sub-mode to use (wideband only)

4.1.2.32 #define SPEEX_SET_MODE 6

Set sub-mode to use

4.1.2.33 #define SPEEX_SET_PF 0

Equivalent to SPEEX_SET_ENH

4.1.2.34 #define SPEEX_SET_QUALITY 4

Set quality value

4.1.2.35 #define SPEEX_SET_SAMPLING_RATE 24

Set sampling rate used in bit-rate computation

4.1.2.36 #define SPEEX_SET_USER_HANDLER 22

Define a handler function for in-band user-defined request

4.1.2.37 #define SPEEX_SET_VAD 30

Set VAD status (1 for on, 0 for off)

4.1.2.38 #define SPEEX_SET_VBR 12

Set VBR on (1) or off (0)

4.1.2.39 #define SPEEX_SET_VBR_QUALITY 14

Set quality value for VBR encoding (0-10)

4.1.2.40 #define SPEEX_SUBMODE_BITS_PER_FRAME 1

Query the size of an encoded frame for a particular sub-mode

4.1.3 Typedef Documentation

4.1.3.1 typedef int(* decode_func)(void *state, SpeexBits *bits, float *out)

Main decoding function

4.1.3.2 typedef int(* decoder_ctl_func)(void *state, int request, void *ptr)

Function for controlling the decoder options

4.1.3.3 typedef void(* decoder_destroy_func)(void *st)

Decoder state destruction function

4.1.3.4 typedef void*(* decoder_init_func)(struct SpeexMode *mode)

Decoder state initialization function

4.1.3.5 typedef int(* encode_func)(void *state, float *in, SpeexBits *bits)

Main encoding function

4.1.3.6 typedef int(* encoder_ctl_func)(void *state, int request, void *ptr)

Function for controlling the encoder options

4.1.3.7 typedef void(* encoder_destroy_func)(void *st)

Encoder state destruction function

4.1.3.8 typedef void*(* encoder_init_func)(struct SpeexMode *mode)

Encoder state initialization function

4.1.3.9 typedef int(* mode_query_func)(void *mode, int request, void *ptr)

Query function for a mode

4.1.3.10 typedef struct SpeexMode SpeexMode

Struct defining a Speex mode

4.1.4 Function Documentation

4.1.4.1 int speex_decode (void * *state*, SpeexBits * *bits*, float * *out*)

Uses an existing decoder state to decode one frame of speech from bit-stream bits. The output speech is saved written to out.

Parameters:

state Decoder state

bits Bit-stream from which to decode the frame (NULL if the packet was lost)

out Where to write the decoded frame

Returns:

return status (0 for no error, -1 for end of stream, -2 other)

4.1.4.2 int speex_decoder_ctl (void * *state*, int *request*, void * *ptr*)

Used like the ioctl function to control the encoder parameters

Parameters:

state Decoder state

request ioctl-type request (one of the SPEEX_* macros)

ptr Data exchanged to-from function

Returns:

0 for no error, 1 if a terminator is reached, 2 for another error

4.1.4.3 void speex_decoder_destroy (void * *state*)

Frees all resources associated to an existing decoder state.

Parameters:

state State to be destroyed

4.1.4.4 void* speex_decoder_init (SpeexMode * *mode*)

Returns a handle to a newly created decoder state structure. For now, the mode argument can be &nb_mode or &wb_mode . In the future, more modes may be added. Note that for now if you have more than one channels to decode, you need one state per channel.

Parameters:

mode Speex mode (one of speex_nb_mode or speex_wb_mode)

Returns:

A newly created decoder state

4.1.4.5 int speex_encode (void * *state*, float * *in*, SpeexBits * *bits*)

Uses an existing encoder state to encode one frame of speech pointed to by ”in”. The encoded bit-stream is saved in ”bits”.

Parameters:

state Encoder state

in Frame that will be encoded with a +-2^16 range

bits Bit-stream where the data will be written

4.1.4.6 int speex_encoder_ctl (void * *state*, int *request*, void * *ptr*)

Used like the ioctl function to control the encoder parameters

Parameters:

state Encoder state

request ioctl-type request (one of the SPEEX_* macros)

ptr Data exchanged to-from function

Returns:
0 if frame needs not be transmitted (DTX only), 1 otherwise

4.1.4.7 void speex_encoder_destroy (void * *state*)

Frees all resources associated to an existing Speex encoder state.

Parameters:
state Encoder state to be destroyed

4.1.4.8 void* speex_encoder_init (SpeexMode * *mode*)

Returns a handle to a newly created Speex encoder state structure. For now, the "mode" argument can be &nb_mode or &wb_mode . In the future, more modes may be added. Note that for now if you have more than one channels to encode, you need one state per channel.

Parameters:
mode The mode to use (either speex_nb_mode or speex_wb.mode)

Returns:
A newly created encoder

4.1.4.9 int speex_mode_query (SpeexMode * *mode*, int *request*, void * *ptr*)

Query function for mode information

Parameters:
mode Speex mode

request ioctl-type request (one of the SPEEX_* macros)

ptr Data exchanged to-from function

4.1.5 Variable Documentation

4.1.5.1 SpeexMode* speex_mode_list[SPEEX_NB_MODES]

List of all modes available

4.1.5.2 SpeexMode speex_nb_mode

Default narrowband mode

4.1.5.3 SpeexMode speex_uwb_mode

Default "ultra-wideband" mode

4.1.5.4 SpeexMode speex_wb_mode

Default wideband mode

4.2 speex_bits.h File Reference

Handles bit packing/unpacking.

Compounds

- struct SpeexBits

Defines

- #define MAX_BYTES_PER_FRAME 2000

Typedefs

- typedef SpeexBits SpeexBits

Functions

- void speex_bits_init (SpeexBits *bits)
- void speex_bits_init_buffer (SpeexBits *bits, void *buff, int buf_size)
- void speex_bits_destroy (SpeexBits *bits)
- void speex_bits_reset (SpeexBits *bits)
- void speex_bits_rewind (SpeexBits *bits)
- void speex_bits_read_from (SpeexBits *bits, char *bytes, int len)
- void speex_bits_read_whole_bytes (SpeexBits *bits, char *bytes, int len)
- int speex_bits_write (SpeexBits *bits, char *bytes, int max_len)
- int speex_bits_write_whole_bytes (SpeexBits *bits, char *bytes, int max_len)
- void speex_bits_pack (SpeexBits *bits, int data, int nbBits)
- int speex_bits_unpack_signed (SpeexBits *bits, int nbBits)
- unsigned int speex_bits_unpack_unsigned (SpeexBits *bits, int nbBits)
- int speex_bits_nbytes (SpeexBits *bits)
- unsigned int speex_bits_peek_unsigned (SpeexBits *bits, int nbBits)
- int speex_bits_peek (SpeexBits *bits)
- void speex_bits_advance (SpeexBits *bits, int n)
- int speex_bits_remaining (SpeexBits *bits)

4.2.1 Detailed Description

Handles bit packing/unpacking.

4.2.2 Define Documentation

4.2.2.1 #define MAX_BYTES_PER_FRAME 2000

Maximum size of the bit-stream (for fixed-size allocation)

4.2.3 Typedef Documentation

4.2.3.1 typedef struct SpeexBits SpeexBits

Bit-packing data structure representing (part of) a bit-stream.

4.2.4 Function Documentation

4.2.4.1 void speex_bits_advance (SpeexBits * *bits*, int *n*)

Advances the position of the "bit cursor" in the stream

Parameters:

bits Bit-stream to operate on

n Number of bits to advance

4.2.4.2 void speex_bits_destroy (SpeexBits * *bits*)

Frees all resources associated to a SpeexBits struct. Right now this does nothing since no resources are allocated, but this could change in the future.

4.2.4.3 void speex_bits_init (SpeexBits * *bits*)

Initializes and allocates resources for a SpeexBits struct

4.2.4.4 void speex_bits_init_buffer (SpeexBits * *bits*, void * *buff*, int *buf_size*)

Initializes SpeexBits struct using a pre-allocated buffer

4.2.4.5 int speex_bits_nbytes (SpeexBits * *bits*)

Returns the number of bytes in the bit-stream, including the last one even if it is not "full"

Parameters:

bits Bit-stream to operate on

Returns:

Number of bytes in the stream

4.2.4.6 void speex_bits_pack (SpeexBits * *bits*, int *data*, int *nbBits*)

Append bits to the bit-stream

Parameters:

bits Bit-stream to operate on

data Value to append as integer

nbBits number of bits to consider in "data"

4.2.4.7 int speex_bits_peek (SpeexBits * *bits*)

Get the value of the next bit in the stream, without modifying the ”cursor” position

Parameters:
bits Bit-stream to operate on

4.2.4.8 unsigned int speex_bits_peek_unsigned (SpeexBits * *bits*, int *nbBits*)

Same as speex_bits_unpack_unsigned, but without modifying the cursor position

4.2.4.9 void speex_bits_read_from (SpeexBits * *bits*, char * *bytes*, int *len*)

Initializes the bit-stream from the data in an area of memory

4.2.4.10 void speex_bits_read_whole_bytes (SpeexBits * *bits*, char * *bytes*, int *len*)

Append bytes to the bit-stream

Parameters:
bits Bit-stream to operate on
bytes pointer to the bytes what will be appended
len Number of bytes of append

4.2.4.11 int speex_bits_remaining (SpeexBits * *bits*)

Returns the number of bits remaining to be read in a stream

Parameters:
bits Bit-stream to operate on

4.2.4.12 void speex_bits_reset (SpeexBits * *bits*)

Resets bits to initial value (just after initialization, erasing content)

4.2.4.13 void speex_bits_rewind (SpeexBits * *bits*)

Rewind the bit-stream to the beginning (ready for read) without erasing the content

4.2.4.14 int speex_bits_unpack_signed (SpeexBits * *bits*, int *nbBits*)

Interpret the next bits in the bit-stream as a signed integer

Parameters:
bits Bit-stream to operate on
nbBits Number of bits to interpret

Returns:
A signed integer represented by the bits read

4.2.4.15 unsigned int speex_bits_unpack_unsigned (SpeexBits * *bits*, int *nbBits*)

Interpret the next bits in the bit-stream as an unsigned integer

Parameters:

bits Bit-stream to operate on

nbBits Number of bits to interpret

Returns:

An unsigned integer represented by the bits read

4.2.4.16 int speex_bits_write (SpeexBits * *bits*, char * *bytes*, int *max_len*)

Write the content of a bit-stream to an area of memory

4.2.4.17 int speex_bits_write_whole_bytes (SpeexBits * *bits*, char * *bytes*, int *max_len*)

Like speex_bits_write, but writes only the complete bytes in the stream. Also removes the written bytes from the stream

4.3 speex_callbacks.h File Reference

Describes callback handling and in-band signalling.

```
#include "speex.h"
```

Compounds

- struct SpeexCallback

Defines

- #define SPEEX_MAX_CALLBACKS 16
- #define SPEEX_INBAND_ENH_REQUEST 0
- #define SPEEX_INBAND_RESERVED1 1
- #define SPEEX_INBAND_MODE_REQUEST 2
- #define SPEEX_INBAND_LOW_MODE_REQUEST 3
- #define SPEEX_INBAND_HIGH_MODE_REQUEST 4
- #define SPEEX_INBAND_VBR_QUALITY_REQUEST 5
- #define SPEEX_INBAND_ACKNOWLEDGE_REQUEST 6
- #define SPEEX_INBAND_VBR_REQUEST 7
- #define SPEEX_INBAND_CHAR 8
- #define SPEEX_INBAND_STEREO 9
- #define SPEEX_INBAND_MAX_BITRATE 10
- #define SPEEX_INBAND_ACKNOWLEDGE 12

Typedefs

- typedef int(* speex_callback_func)(SpeexBits *bits, void *state, void *data)
- typedef SpeexCallback SpeexCallback

Functions

- int speex_inband_handler (SpeexBits *bits, SpeexCallback *callback_list, void *state)
- int speex_std_mode_request_handler (SpeexBits *bits, void *state, void *data)
- int speex_std_high_mode_request_handler (SpeexBits *bits, void *state, void *data)
- int speex_std_char_handler (SpeexBits *bits, void *state, void *data)
- int speex_default_user_handler (SpeexBits *bits, void *state, void *data)
- int **speex_std_low_mode_request_handler** (SpeexBits *bits, void *state, void *data)
- int **speex_std_vbr_request_handler** (SpeexBits *bits, void *state, void *data)
- int **speex_std_enh_request_handler** (SpeexBits *bits, void *state, void *data)
- int **speex_std_vbr_quality_request_handler** (SpeexBits *bits, void *state, void *data)

4.3.1 Detailed Description

Describes callback handling and in-band signalling.

4.3.2 Define Documentation

4.3.2.1 #define SPEEX_INBAND_ACKNOWLEDGE 12

Acknowledge packet reception

4.3.2.2 #define SPEEX_INBAND_ACKNOWLEDGE_REQUEST 6

Request to be sent acknowledge

4.3.2.3 #define SPEEX_INBAND_CHAR 8

Send a character in-band

4.3.2.4 #define SPEEX_INBAND_ENH_REQUEST 0

Request for perceptual enhancement (1 for on, 0 for off)

4.3.2.5 #define SPEEX_INBAND_HIGH_MODE_REQUEST 4

Request for a high mode change

4.3.2.6 #define SPEEX_INBAND_LOW_MODE_REQUEST 3

Request for a low mode change

4.3.2.7 #define SPEEX_INBAND_MAX_BITRATE 10

Transmit max bit-rate allowed

4.3.2.8 #define SPEEX_INBAND_MODE_REQUEST 2

Request for a mode change

4.3.2.9 #define SPEEX_INBAND_RESERVED1 1

Reserved

4.3.2.10 #define SPEEX_INBAND_STEREO 9

Intensity stereo information

4.3.2.11 #define SPEEX_INBAND_VBR_QUALITY_REQUEST 5

Request for VBR (1 on, 0 off)

4.3.2.12 #define SPEEX_INBAND_VBR_REQUEST 7

Request for VBR (1 for on, 0 for off)

4.3.2.13 #define SPEEX_MAX_CALLBACKS 16

Total number of callbacks

4.3.3 Typedef Documentation

4.3.3.1 typedef int(* speex_callback_func)(SpeexBits *bits, void *state, void *data)

Callback function type

4.3.3.2 typedef struct SpeexCallback SpeexCallback

Callback information

4.3.4 Function Documentation

4.3.4.1 int speex_default_user_handler (SpeexBits * *bits*, void * *state*, void * *data*)

Default handler for user-defined requests: in this case, just ignore

4.3.4.2 int speex_inband_handler (SpeexBits * *bits*, SpeexCallback * *callback_list*, void * *state*)

Handle in-band request

4.3.4.3 int speex_std_char_handler (SpeexBits * *bits*, void * *state*, void * *data*)

Standard handler for in-band characters (write to stderr)

4.3.4.4 int speex_std_high_mode_request_handler (SpeexBits * *bits*, void * *state*, void * *data*)

Standard handler for high mode request (change high mode, no questions asked)

4.3.4.5 int speex_std_mode_request_handler (SpeexBits * *bits*, void * *state*, void * *data*)

Standard handler for mode request (change mode, no questions asked)

4.4 speex_header.h File Reference

Describes the Speex header.

Compounds

- struct SpeexHeader

Defines

- #define SPEEX_HEADER_VERSION_LENGTH 20

Typedefs

- typedef SpeexHeader SpeexHeader

Functions

- void speex_init_header (SpeexHeader *header, int rate, int nb_channels, struct SpeexMode *m)
- char * speex_header_to_packet (SpeexHeader *header, int *size)
- SpeexHeader * speex_packet_to_header (char *packet, int size)

4.4.1 Detailed Description

Describes the Speex header.

4.4.2 Define Documentation

4.4.2.1 #define SPEEX_HEADER_VERSION_LENGTH 20

Maximum number of characters for encoding the Speex version number in the header

4.4.3 Typedef Documentation

4.4.3.1 typedef struct SpeexHeader SpeexHeader

Speex header info for file-based formats

4.4.4 Function Documentation

4.4.4.1 char* speex_header_to_packet (SpeexHeader * *header*, int * *size*)

Creates the header packet from the header itself (mostly involves endianness conversion)

4.4.4.2 void speex_init_header (SpeexHeader * *header*, int *rate*, int *nb_channels*, struct SpeexMode * *m*)

Initializes a SpeexHeader using basic information

4.4.4.3 SpeexHeader* speex_packet_to_header (char * *packet*, int *size*)

Creates a SpeexHeader from a packet

4.5 speex_stereo.h File Reference

Describes the handling for intensity stereo.

`#include "speex_bits.h"`

Compounds

- struct SpeexStereoState

Defines

- #define SPEEX_STEREO_STATE_INIT {1,.5,1,1}

Typedefs

- typedef SpeexStereoState SpeexStereoState

Functions

- void speex_encode_stereo (float *data, int frame_size, SpeexBits *bits)
- void speex_decode_stereo (float *data, int frame_size, SpeexStereoState *stereo)
- int speex_std_stereo_request_handler (SpeexBits *bits, void *state, void *data)

4.5.1 Detailed Description

Describes the handling for intensity stereo.

4.5.2 Define Documentation

4.5.2.1 #define SPEEX_STEREO_STATE_INIT {1,.5,1,1}

Initialization value for a stereo state

4.5.3 Typedef Documentation

4.5.3.1 typedef struct SpeexStereoState SpeexStereoState

State used for decoding (intensity) stereo information

4.5.4 Function Documentation

4.5.4.1 void speex_decode_stereo (float * *data*, int *frame_size*, SpeexStereoState * *stereo*)

Transforms a mono frame into a stereo frame using intensity stereo info

4.5.4.2 void speex_encode_stereo (float * *data*, int *frame_size*, SpeexBits * *bits*)

Transforms a stereo frame into a mono frame and stores intensity stereo info in 'bits'

4.5.4.3 int speex_std_stereo_request_handler (SpeexBits * *bits*, void * *state*, void * *data*)

Callback handler for intensity stereo info

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